

AKADEMATE

UI and frontend architecture description

Seven product surfaces on Next.js 15 App Router. Tokens as CSS variables. shadow new-york primitives. Shells per surface. Catalog order for every new screen.

Legal owner	BRIK64 Inc. (Delaware)
Identification	AKA-ARCH-UI-001
Revision / issue	A · 2026-08-30
Classification	Internal · Confidential
Sheet	1 / 46

0.1 Identification

Field	Value
Legal owner	BRIK64 Inc., Delaware, United States. Foundry. Initial legal owner.
Identification number	AKA-ARCH-UI-001
Revision index	A
Date of issue	2026-08-30
Effective	On last approval signature
Next review	2026-11-30
Language	en (ISO 639)
Document type	Architecture description
Product	AKADEMATE
System of interest	AKADEMATE UI and frontend across marketing, academy, campus, ops and CMS
Classification	Internal · Confidential
Paper size	A4
Normative sources	ISO/IEC/IEEE 42010 · ISO 7200 · arc42 · ADR-0004 · ACADEIMATE SPEC v1.5
Machine ingest	AKA-ARCH-UI-001.ingest.json · AKA-ARCH-UI-001.AGENT.md · diagrams/*.mmd · Appendix A

0.1b Approvals

Role	Organisation	Name	Date
Creator / Foundry	BRIK64 Inc.		
Architect	AKADEMATE		
Product	AKADEMATE		
Design	AKADEMATE		
Approval	BRIK64 Inc. / AKADEMATE		

Creator / Foundry
 BRIK64 Inc.

SignatureDate

Architect
 AKADEMATE

SignatureDate

Product
 AKADEMATE

SignatureDate

Design
 AKADEMATE

SignatureDate

Approval
 BRIK64 Inc. / AKADEMATE

SignatureDate

0.2 Stakeholders and concerns

Stakeholder	Concern
Academy staff	Dense, keyboard-reachable dashboard. One chrome. Tokens follow the academy brand.
Learners and teachers	Campus and public pages that read as the academy, with a clear login.
Platform ops	A separate ops shell. Dark default. No academy sidebar mix-in.
Foundry / design	One catalog, one token pipeline, WCAG AA, no painted-blank screens.
Implementing agent	Codes, Mermaid source, and a prompt that names slugs before paint.

0.3 Viewpoints

Viewpoint	View in this description
Context	Seven surfaces. Two public webs. Live product vs scaffold.
Functional	Shells, primitives, catalog order, auth composition.
Runtime	Token injection, class-based light/dark, compose sequence.
Deployment	App packages, design-system pages, Storybook.
Quality	AA contrast, focus, keyboard, motion policy.

Companion description: AKA-ARCH-SAAS-001 (Cloudflare runtime). This document owns presentation.

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1 Introduction and goals

AKADEMATE presents one product through several windows: marketing, academy dashboard, tenant public site, campus, platform ops, access hub, and CMS admin.

The goal is a single visual language. Tokens in CSS variables. Primitives from shadcn new-york. A shell per surface. An implementing agent reads codes and Mermaid the same way a human reads the sheet.

Quality goals

- WCAG 2.1 AA on text and controls.
- Visible focus and a complete keyboard path.
- Tenant brand through variables, one primitive catalog.
- Every new screen names a catalog slug before paint.

2 Constraints

- Next.js 15 App Router, React 19, TypeScript.
- Tailwind v4 with colors in `theme.colors` and PostCSS @tailwindcss/postcss.
- shadcn/ui new-york, CSS variables, Lucide as the icon library of record.
- Multitenant brand: CSS variables, never a second component tree per tenant.
- Dashboard chrome stays on SaaS tokens. Tenant public may apply a host overlay through variables.
- Product chrome animation: CSS, Radix, tw-animate. Marketing motion: Motion library by slug.

3 Context and scope

System of interest: presentation for AKADEMATE. Runtime compute and data live in AKA-ARCH-SAAS-001.

In scope: surfaces, tokens, primitives, shells, theming, auth screens, catalog order, access, design-system delivery.

Live product UI: SURF-WEB, SURF-DASH, SURF-PUBLIC, SURF-CAMPUS, SURF-OPS, SURF-CMS. SURF-PORTAL is the access hub. `apps/campus` and `apps/ops` are scaffolds beside the live surfaces.

Figure 1.

4 Solution strategy

1. Own each window as a surface code (SURF-*).
2. Feed color and type from TOK-* . Runtime brand is TenantBrandingProvider.
3. Compose pages from CAT-REPO then CAT-SHADCN. Motion by slug only.
4. Give each surface one shell. Dashboard and tenant public do not share html.
5. Lift primitives into @akademate/ui while the Payload-config alias keeps resolving.
6. Complete SURF-WEB on Tailwind v4 `theme.colors`.

5.1 Building blocks: surfaces

Seven windows. One product. Figure 1.

Code	Location	Role
SURF-WEB	apps/web	SaaS marketing
SURF-DASH	apps/tenant-admin (app) / (dashboard)	Academy dashboard
SURF-PUBLIC	apps/tenant-admin (public)	Tenant public site
SURF-CAMPUS	apps/tenant-admin (app) / campus	Academy campus
SURF-OPS	apps/admin-client	Platform ops
SURF-PORTAL	apps/portal	Access hub
SURF-CMS	Payload admin	CMS chrome

SURF-WEB is SaaS marketing on akademate.com. SURF-PUBLIC is the academy site on the tenant host. Figure 2.

5.2 Building blocks: tokens

Three-layer tokens: primitive, semantic, tenant. Tailwind v4 maps them through TOK-THEME-V4. Figure 3.

Code	Source	Role
TOK-BASE	packages/ui/src/tokens/base.css	Primitive palette and type
TOK-SEM	packages/ui/src/tokens/semantic.css	Purpose aliases
TOK-TENANT	packages/ui/src/tokens/tenant.css	Brand accent
TOK-THEME-V4	@theme inline	Tailwind v4 color map
TOK-RUNTIME	TenantBrandingProvider	Injects --primary and --sidebar

Locked defaults: Manrope in TOK-BASE, runtime primary #0066CC, sidebar #0F2440. Light and dark as .light / .dark on html.

5.3 Building blocks: primitives

Canonical catalog: shadcn new-york in tenant-admin `components.json`, `cssVariables: true`, `iconLibrary: lucide`. Dashboard imports `@payload-config/components/ui`.

Figure 4.

Stable slugs: accordion, alert, alert-dialog, avatar, badge, breadcrumb, button, card, checkbox, dialog, dropdown-menu, input, label, select, separator, sheet, sidebar, skeleton, switch, table, tabs, textarea, tooltip.

`@akademate/ui` exports tokens and Pill. The target is the same slugs in that package. SURF-WEB, SURF-PORTAL and SURF-OPS keep smaller local trees until the lift lands.

6.1 Runtime: compose

A route selects a shell. The shell applies tokens. The page composes primitives. Figure 11.

1. Layout (SHELL-*) wraps the segment.
2. ThemeProvider sets `light` or `dark` on `html`.
3. TenantBrandingProvider writes `--primary`, `--sidebar`, `--brand` from academy color.
4. The page imports `button`, `card`, `dialog`, `table` from the catalog alias.

Forms: React Hook Form and zod on dashboard flows.

6.2 Runtime: theming

Three knobs, one tree. Figure 7.

- Class: `light` / `dark` on `html`. Storage key `akademate-theme` on the academy app. Ops uses `akademate-ops-theme`, `dark` default.
- Variables: `--primary`, `--sidebar`, `--brand`, `--ring` from TOK-RUNTIME.
- SURF-WEB may read an `akademate_theme` cookie into the same variables.

Tenant public injects `--brand` from the academy primary color. Overlay type and footer copy stay variables and content, not a fork of button or card.

7.1 Deployment: apps

Surface	Package	Notes
SURF-WEB	apps/web	Marketing Worker. Completes Tailwind v4 cutover.
SURF-DASH, PUBLIC, CAMPUS, CMS	apps/tenant-admin	Primary product UI. Payload admin in the same app.
SURF-OPS	apps/admin-client	Platform ops.
SURF-PORTAL	apps/portal	Access hub.

Scaffolds `apps/campus` and `apps/ops` remain beside the live surfaces until callers move to SURF-CAMPUS and SURF-OPS.

7.2 Deployment: design system

Delivery surfaces, not a second token source. Figure 10.

- SURF-WEB /`design-system` and template sandbox.
- SURF-DASH /`design-system` playground.
- Storybook on tenant-admin at port 6006: foundations, patterns, akademate blocks.
- Vendor kit copy-in: templates restyle to TOK-* at the moment of install.

8.1 Shells

One shell per window. Figures 5 and 6.

Code	Implementation	Surface
SHELL-DASH	AppSidebar + SidebarProvider	Dashboard
SHELL-PUBLIC	Public header and footer	Tenant public
SHELL-OPS	OpsSidebar	Platform ops
SHELL-CAMPUS	CampusNavbar	Campus
SHELL-AUTH	Auth canvas	Staff login
SHELL-MKT	Marketing header and footer	SURF-WEB

SHELL-DASH: AppSidebar, SidebarProvider, SidebarInset, DashboardFooter, CommandPalette, NotificationBell, ThemeToggle. Canvas token --dashboard-canvas.

8.2 Auth

Every live login is Card, Input, Button. Figure 8.

- SURF-DASH: /auth/login, signup, forgot-password, accept-invite. SHELL-AUTH canvas.
- SURF-CAMPUS: /campus/login.
- SURF-OPS: /login.
- SURF-WEB: /login continues to the academy app login. /registro is the lead form.
- SURF-CMS: Payload admin login.

8.3 Catalog and motion

Do not invent a hero, pricing, sidebar, form, empty state, or chat. Figure 9.

Code	Source	When
CAT-REPO	components/ui in the app tree	First
CAT-SHADCN	shadcn new-york official slug	Second
CAT-MOTION	Magic UI or Aceternity by slug	Marketing motion
CAT-EXT	One 21st.dev or ReUI block	If the pattern is missing

Chat UI is the official shadcn block. Product chrome motion is CSS, Radix, tw-animate. Marketing motion is the Motion library by slug.

8.4 Access

Figure 12.

- WCAG 2.1 AA contrast on text and UI components.
- Visible focus ring on every control. Keyboard path through sidebar, dialog, sheet, select, tabs.
- Semantic headings, labels, and live regions (sonner toasts).
- Hit targets at least 44 by 44 CSS pixels on public and campus.
- Decorative icons: `aria-hidden`. Icon-only buttons: accessible name.

9 Architecture decisions

Code	Decision
ADR-U1	Product UIs are Next.js 15 App Router.
ADR-U2	Color is CSS variables in HSL channels. Tenant branding writes --primary and --sidebar.
ADR-U3	The shadcn new-york catalog lives in tenant-admin. Dashboard imports the Payload-config alias.
ADR-U4	@akademate/ui ships tokens and Pill. Primitives migrate into that package.
ADR-U5	Dashboard chrome reads --sidebar and TenantBranding.
ADR-U6	Tenant public is its own html document, separate from the dashboard shell.
ADR-U7	Platform ops UI is apps/admin-client.
ADR-U8	Campus product routes live under tenant-admin. apps/campus is a scaffold.
ADR-U9	Light and dark are a class on html.
ADR-U10	Compose from catalog order: repo, official shadcn, motion by slug.
ADR-U11	Tailwind v4 with theme.colors and @tailwindcss/postcss. SURF-WEB completes that cutover.
ADR-U12	Motion on product chrome is CSS, Radix, and tw-animate. Marketing motion uses the Motion library by slug.

10 Quality requirements

Quality	Requirement
Accessibility	AA, focus, keyboard, names on icon buttons.
Consistency	One catalog. One token pipeline. Shell per surface.
Tenant brand	Variables only. Dashboard chrome stays SaaS-tokenized.
Performance	next/image, self-hosted fonts, CSS animation on chrome.
Evolvability	Primitive lift into @akademate/ui with the alias intact.

11 Glossary and references

Term	Definition
Surface	A product window with its own shell and routes.
Primitive	A catalog component (button, dialog, sidebar).
Token	A CSS variable for color, type, space, or radius.
Host overlay	Tenant public type, footer and contact channels on top of the same primitives.
Catalog order	Repo, official shadcn, motion by slug, one extension block.

References

- ISO/IEC/IEEE 42010:2022 · ISO 7200:2004 · arc42
- docs/adr/0004-ui-kit.md
- docs/specs/ACADEIMATE_SPEC.md v1.5
- AKA-ARCH-SAAS-001
- ui.shadcn.com new-york

Revision history

Rev	Date	Change
A	2026-08-30	Initial UI and frontend architecture description.

12 Agent ingest

This description ships a machine form so an implementing agent reads the same source a human reads on paper.

Read order

1. `AKA-ARCH-UI-001.ingest.json`: codes, ADRs, surfaces, shells, tokens, catalog, every figure as Mermaid.
2. `diagrams/*.mmd`: figure sources, MIME text/vnd.mermaid.
3. `AKA-ARCH-UI-001.AGENT.md`: implementation prompt.
4. This HTML, or the PDF text layer: ingest sheets, figure atlas, and Appendix A.

Each figure sheet names its source. Example: `FIG-01 · diagrams/01-surfaces.mmd`. Appendix A repeats the Mermaid as selectable text.

12.1 Surface codes

Code	Location	Role
SURF-WEB	apps/web	SaaS marketing
SURF-DASH	apps/tenant-admin (app) / (dashboard)	Academy dashboard
SURF-PUBLIC	apps/tenant-admin (public)	Tenant public site
SURF-CAMPUS	apps/tenant-admin (app) / campus	Academy campus
SURF-OPS	apps/admin-client	Platform ops
SURF-PORTAL	apps/portal	Access hub
SURF-CMS	Payload admin	CMS chrome

Entry codes: INGEST-JSON, INGEST-PROMPT, INGEST-LLMS, FIG-01 to FIG-12, ADR-U1 to ADR-U12.

12.2 Token and shell codes

Tokens

Code	Source	Role
TOK-BASE	packages/ui/src/tokens/base.css	Primitive palette and type
TOK-SEM	packages/ui/src/tokens/semantic.css	Purpose aliases
TOK-TENANT	packages/ui/src/tokens/tenant.css	Brand accent
TOK-THEME-V4	@theme inline	Tailwind v4 color map
TOK-RUNTIME	TenantBrandingProvider	Injects --primary and --sidebar

Shells

Code	Implementation	Surface
SHELL-DASH	AppSidebar + SidebarProvider	Dashboard
SHELL-PUBLIC	Public header and footer	Tenant public
SHELL-OPS	OpsSidebar	Platform ops
SHELL-CAMPUS	CampusNavbar	Campus
SHELL-AUTH	Auth canvas	Staff login
SHELL-MKT	Marketing header and footer	SURF-WEB

12.3 Implementation prompt

INGEST-PROMPT · AKA-ARCH-UI-001.AGENT.md · part 1 of 2

```
# AKA-ARCH-UI-001 agent implementation prompt

You are implementing AKADEMATE UI and frontend from architecture description AKA-ARCH-UI-001 (BRIK64 Inc.).

## Ingest order

1. Read AKA-ARCH-UI-001.ingest.json. Canonical machine form: codes, ADRs, surfaces, shells, tokens, catalog, figures as Mermaid.
2. Read each diagrams/*.mmd named in that file. PNG files are renderings of those sources.
3. Read this file for the execution contract.
4. Read AKA-ARCH-UI-001.html or the PDF text layer for the ingest sheets, the figure atlas, and Appendix A.

Appendix A holds Mermaid as selectable text. Figure sheets name the source path (example: FIG-01 · diagrams/01-surfaces.mmd).

## Execution contract

Apply every ADR-U1 through ADR-U12.

Paint from catalog. Search the repo primitive, then official shadcn slug, then Magic UI or Aceternity by slug for marketing motion. Name Superficie / Catalogo / Slug / Source / Tokens / Salto on every new visual surface.

Tokens flow TOK-BASE to TOK-SEM to TOK-TENANT to TOK-THEME-V4. Runtime accent is TOK-RUNTIME
TenantBrandingProvider writing --primary, --sidebar, --brand. Default primary #0066CC. Sidebar default #0F2440.

SURF-DASH uses SHELL-DASH (AppSidebar, SidebarProvider, SidebarInset). SURF-PUBLIC uses SHELL-PUBLIC on its own html document. SURF-OPS uses SHELL-OPS. SURF-CAMPUS uses SHELL-CAMPUS. SURF-WEB uses SHELL-MKT.

Dashboard chrome is SaaS tokens. Tenant public may carry a host overlay (type, footer, contact channels) through CSS variables, not a second primitive catalog.

Canonical primitives today: tenant-admin components/ui, aliased as @payload-config/components/ui. Target: the same slugs in @akademate/ui.

Auth screens compose Card, Input, Button. SURF-WEB /login routes to the academy app login.

Quality: WCAG AA contrast, visible focus, keyboard path, semantic HTML. Product chrome uses CSS and Radix animation. SURF-WEB completes Tailwind v4 theme.colors.
```

12.4 Implementation sequence

INGEST-PROMPT · AKA-ARCH-UI-001.AGENT.md · part 2 of 2

Implementation sequence

1. Map SURF-* to apps and route groups.
2. Wire TOK-* into each live surface. Remove duplicate palettes once the package import is in place.
3. Keep CAT-SHADCN slugs stable (button, card, dialog, sheet, sidebar, input, table, tabs, select, badge, tooltip).
4. Implement SHELL-DASH, SHELL-PUBLIC, SHELL-OPS, SHELL-CAMPUS, SHELL-AUTH, SHELL-MKT.
5. Auth: Card + Input + Button on each live login.
6. Design-system pages and Storybook stay delivery surfaces, not a second source of tokens.
7. Cut SURF-WEB to Tailwind v4 theme.colors and @tailwindcss/postcss.
8. Lift primitives from tenant-admin into @akademate/ui while the alias keeps working.

Done when

Every figure in ingest.json has a matching .mmd. Every SURF, SHELL, TOK, CAT, ADR-U code is present in code or this pack. A new screen names a catalog slug. `verify_print_pdf.py` exits 0 on this PDF.

Figure 1. Product surfaces

FIG-01 · diagrams/01-surfaces.mmd · text/vnd.mermaid · PNG is a rendering of that source

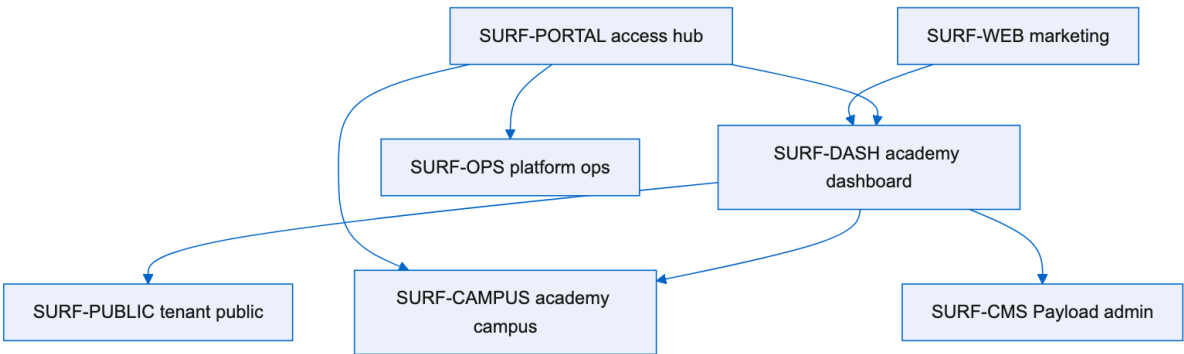


Figure 2. Two public webs

FIG-02 · diagrams/02-dual-public.mmd · text/vnd.mermaid · PNG is a rendering of that source

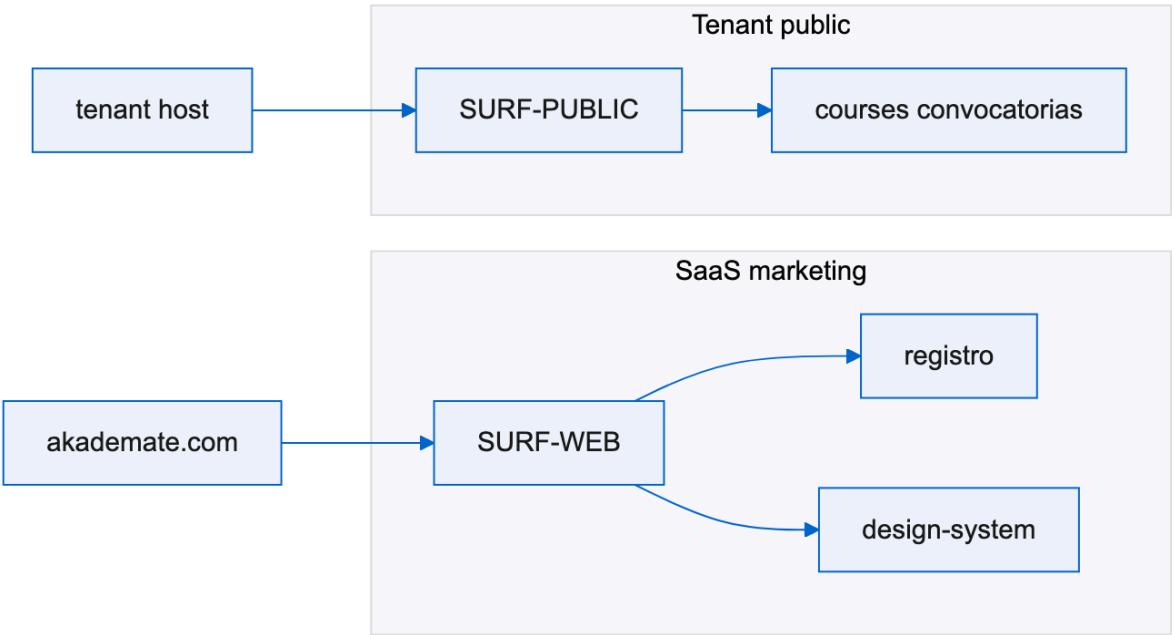


Figure 3. Token pipeline

FIG-03 · diagrams/03-token-pipeline.mmd · text/vnd.mermaid · PNG is a rendering of that source

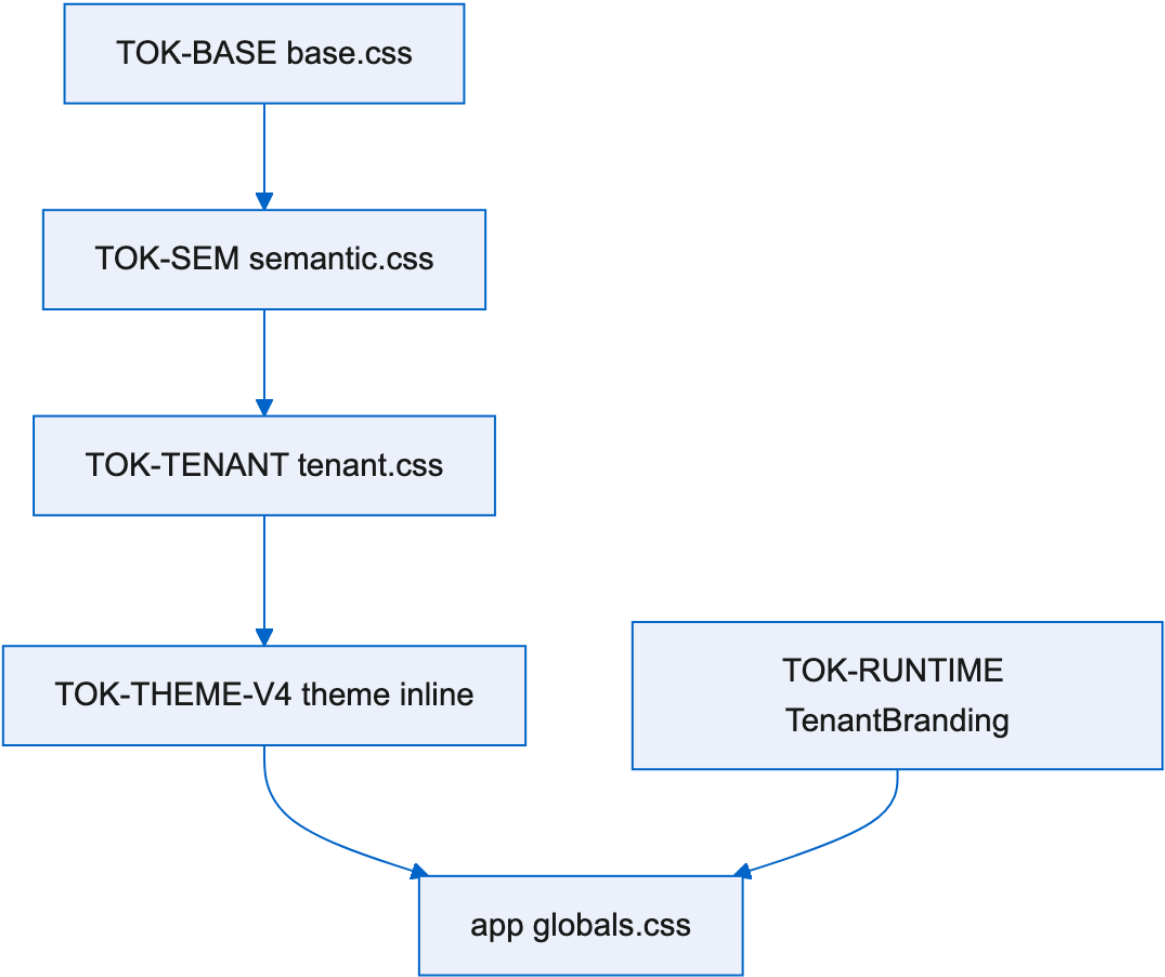


Figure 4. Primitive ownership

FIG-04 · diagrams/04-primitive-ownership.mmd · text/vnd.mermaid · PNG is a rendering of that source

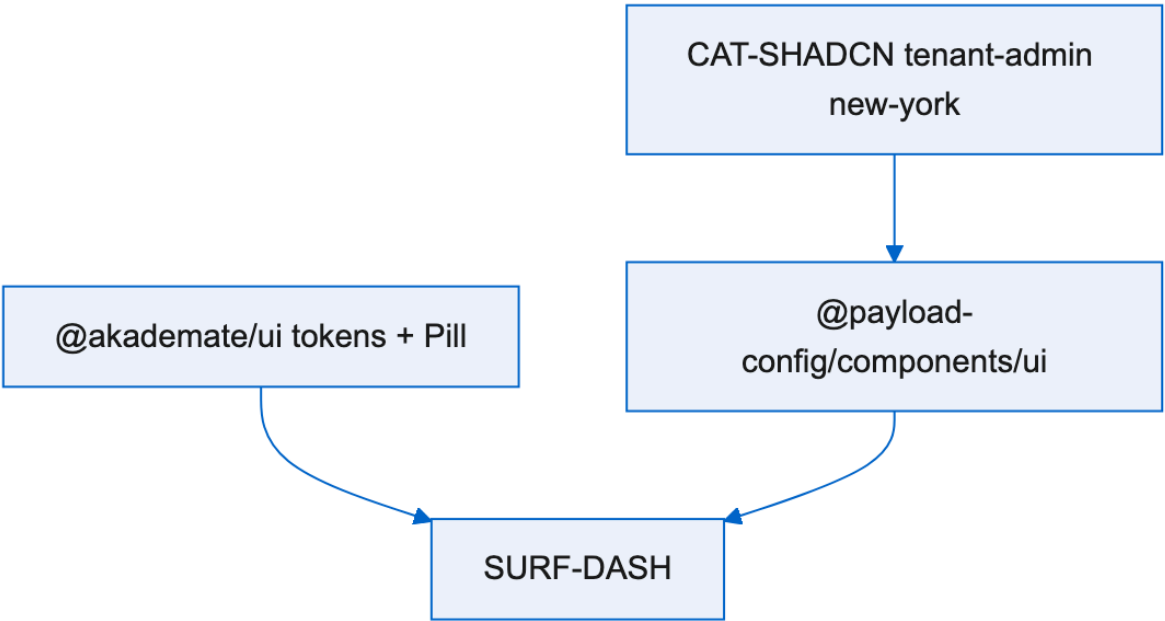


Figure 5. Dashboard shell

FIG-05 · diagrams/05-dashboard-shell.mmd · text/vnd.mermaid · PNG is a rendering of that source

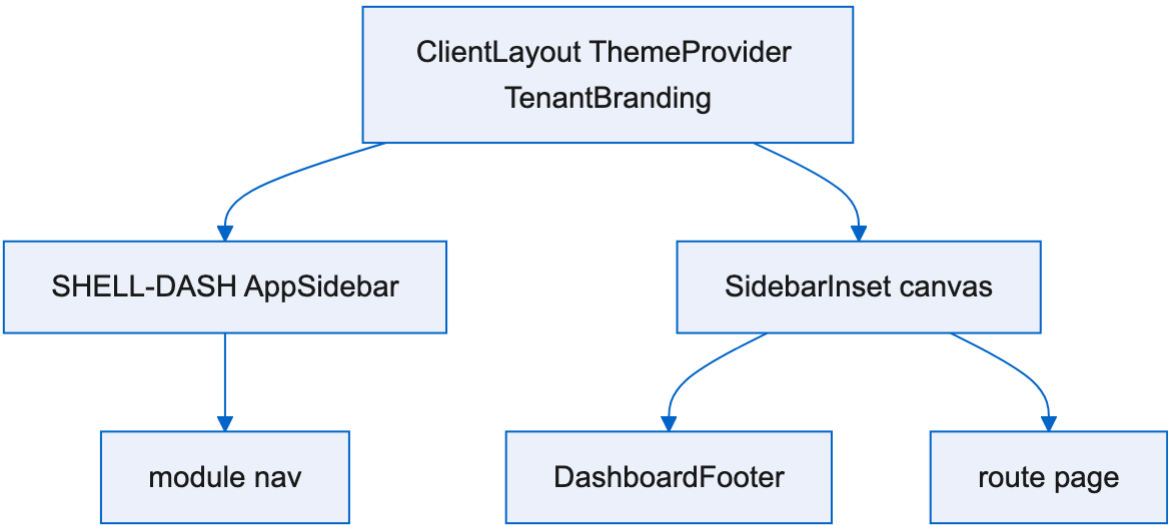


Figure 6. Tenant public shell

FIG-06 · diagrams/06-public-shell.mmd · text/vnd.mermaid · PNG is a rendering of that source

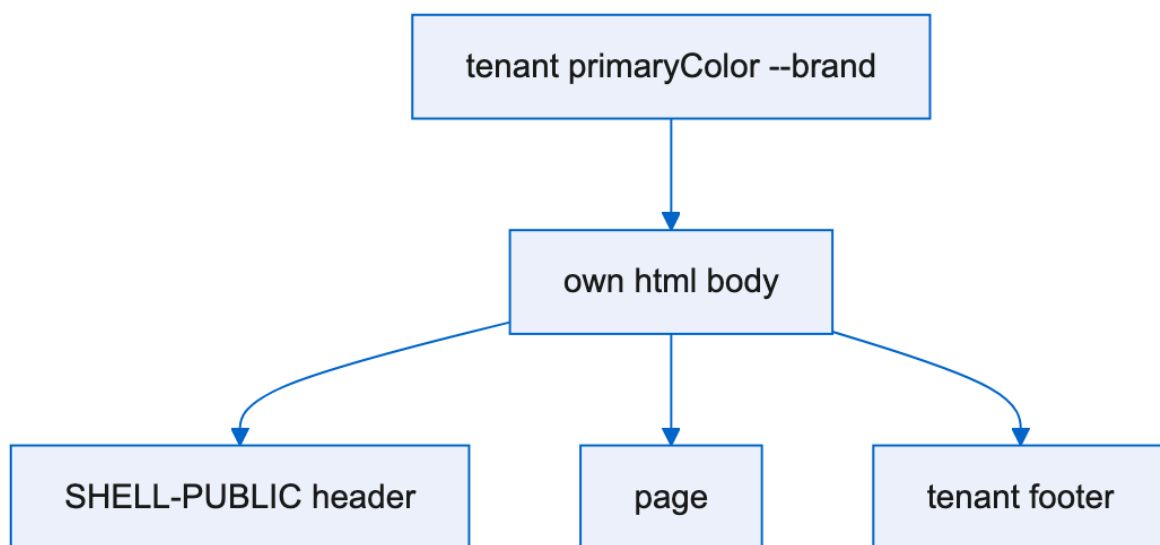


Figure 7. Theming

FIG-07 · diagrams/07-theming.mmd · text/vnd.mermaid · PNG is a rendering of that source

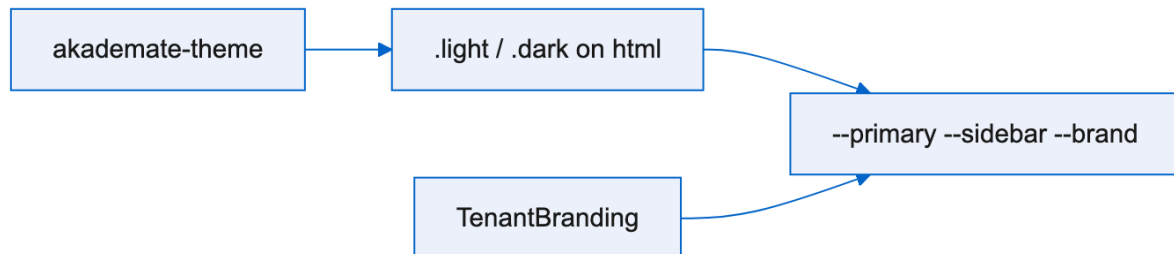


Figure 8. Auth surfaces

FIG-08 · diagrams/08-auth-surfaces.mmd · text/vnd.mermaid · PNG is a rendering of that source

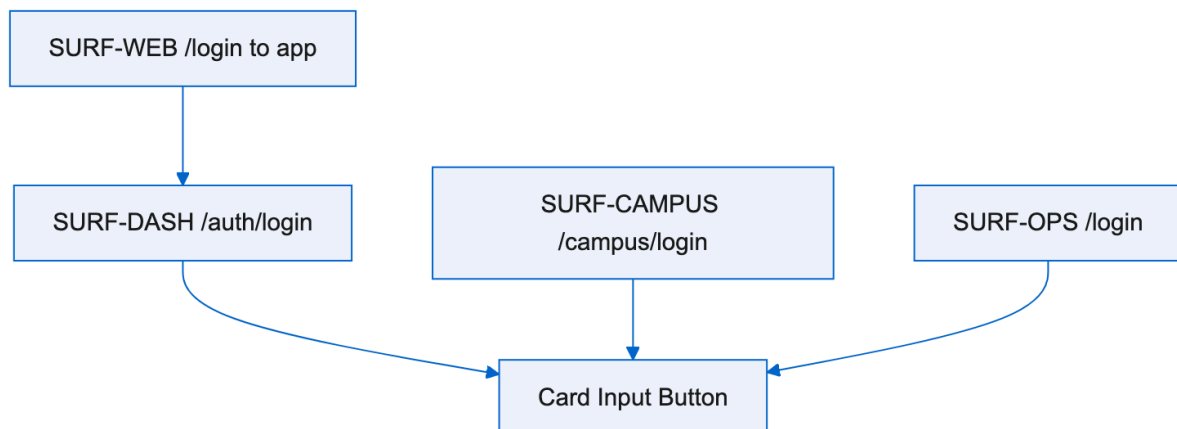


Figure 9. Catalog order

FIG-09 · diagrams/09-catalog-order.mmd · text/vnd.mermaid · PNG is a rendering of that source

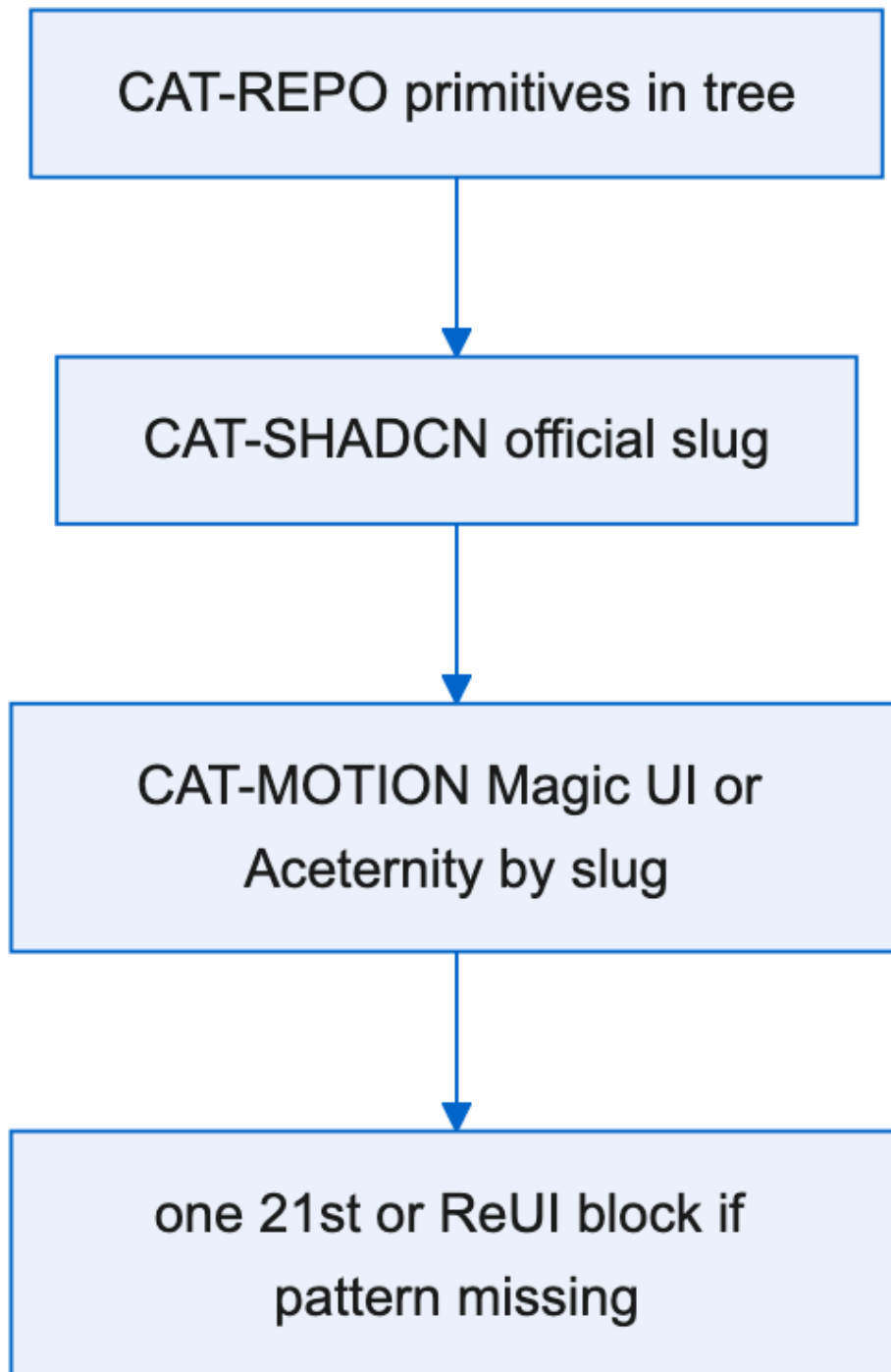


Figure 10. Design-system delivery

FIG-10 · diagrams/10-design-system-delivery.mmd · text/vnd.mermaid · PNG is a rendering of that source

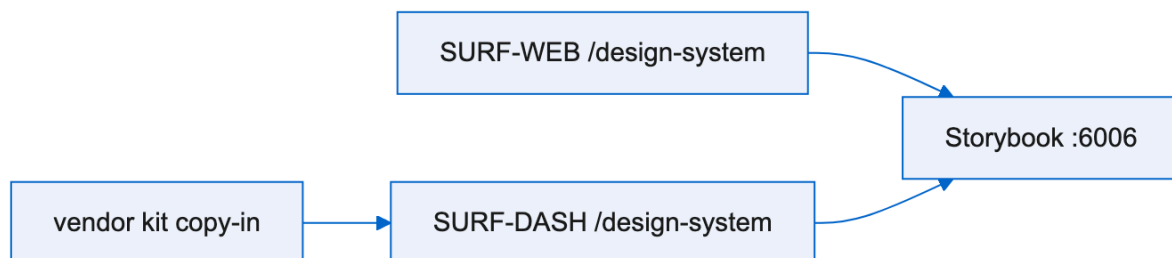


Figure 11. Compose sequence

FIG-11 · diagrams/11-compose-sequence.mmd · text/vnd.mermaid · PNG is a rendering of that source

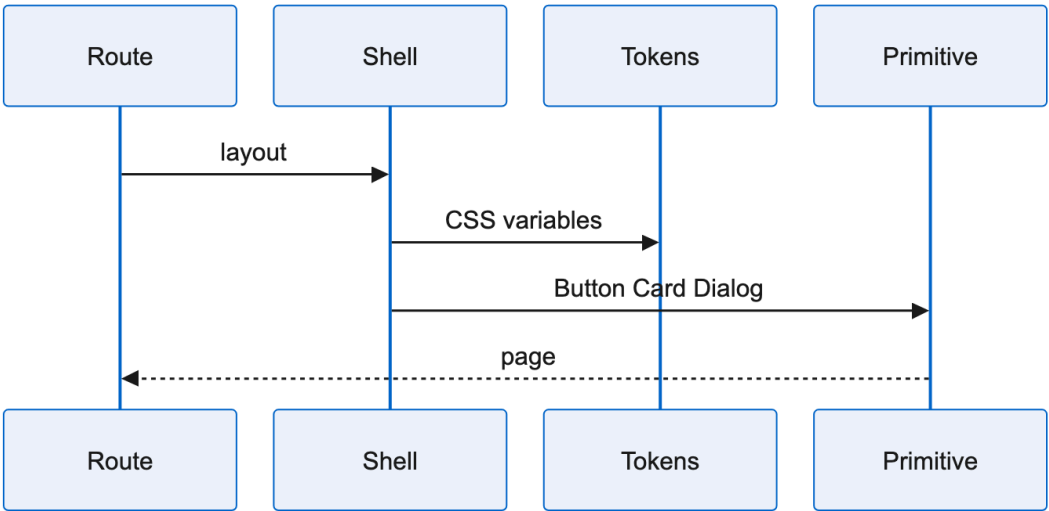
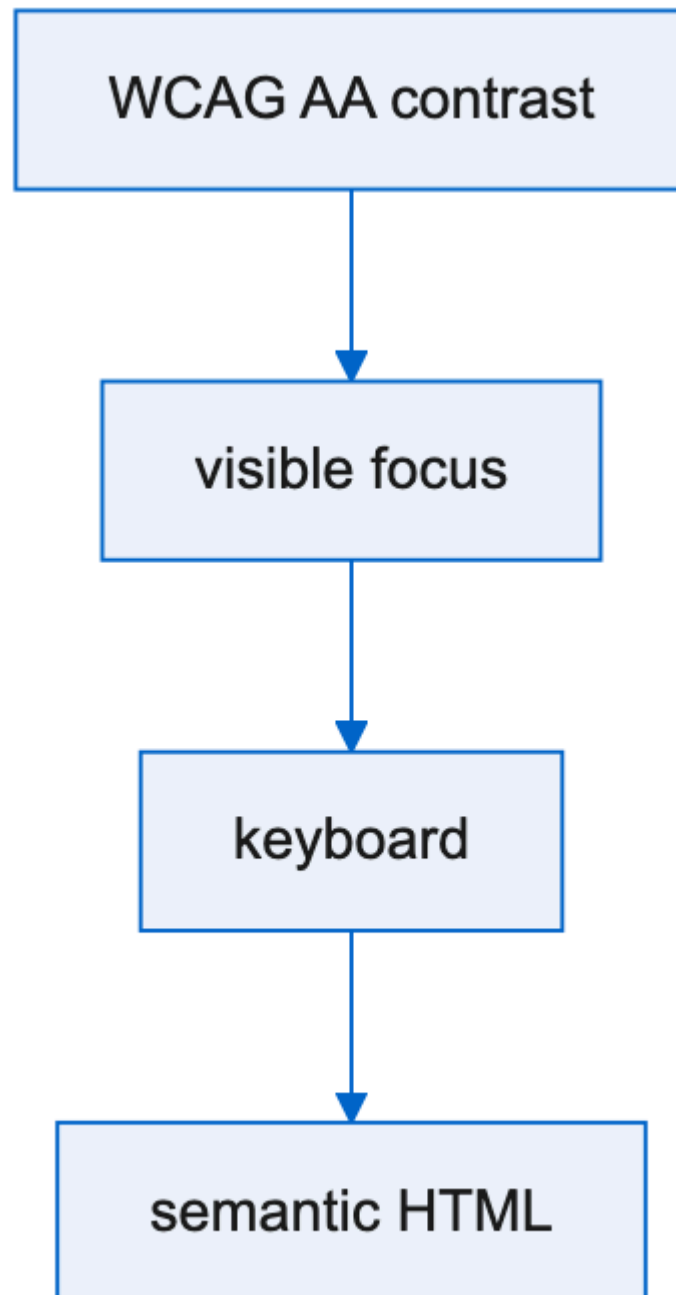


Figure 12. Quality and access

FIG-12 · diagrams/12-quality-ally.mmd · text/vnd.mermaid · PNG is a rendering of that source



A.1 Mermaid source FIG-01 to FIG-03

FIG-01 · diagrams/01-surfaces.mmd · text/vnd.mermaid

```
flowchart TB
  MKT[SURF-WEB marketing]
  DASH[SURF-DASH academy dashboard]
  PUB[SURF-PUBLIC tenant public]
  CAMP[SURF-CAMPUS academy campus]
  OPS[SURF-OPS platform ops]
  PORT[SURF-PORTAL access hub]
  CMS[SURF-CMS Payload admin]
  MKT --> DASH
  PORT --> DASH
  PORT --> CAMP
  PORT --> OPS
  DASH --> PUB
  DASH --> CAMP
  DASH --> CMS
```

FIG-02 · diagrams/02-dual-public.mmd · text/vnd.mermaid

```
flowchart LR
  subgraph SAAS["SaaS marketing"]
    WEB[SURF-WEB]
    REG[registro]
    DS[design-system]
    WEB --> REG
    WEB --> DS
  end
  subgraph TENANT["Tenant public"]
    PUB[SURF-PUBLIC]
    COURSES[courses convocatorias]
    PUB --> COURSES
  end
  HOST[tenant host] --> PUB
  AKA[akademate.com] --> WEB
```

FIG-03 · diagrams/03-token-pipeline.mmd · text/vnd.mermaid

```
flowchart TB
  BASE[TOK-BASE base.css]
  SEM[TOK-SEM semantic.css]
  TEN[TOK-TENANT tenant.css]
  V4[TOK-THEME-V4 theme inline]
  RT[TOK-RUNTIME TenantBranding]
  APP[app globals.css]
  BASE --> SEM
  SEM --> TEN
  TEN --> V4
  V4 --> APP
  RT --> APP
```

A.2 Mermaid source FIG-04 to FIG-06

FIG-04 · diagrams/04-primitive-ownership.mmd · text/vnd.mermaid

```
flowchart TB
  PKG["@akademate/ui tokens + Pill"]
  SHAD[CAT-SHADCN tenant-admin new-york]
  ALIAS["@payload-config/components/ui"]
  DASH[SURF-DASH]
  PKG --> DASH
  SHAD --> ALIAS
  ALIAS --> DASH
```

FIG-05 · diagrams/05-dashboard-shell.mmd · text/vnd.mermaid

```
flowchart TB
  ROOT[ClientLayout ThemeProvider TenantBranding]
  SIDE[SHELL-DASH AppSidebar]
  INSET[SidebarInset canvas]
  FOOT[DashboardFooter]
  ROOT --> SIDE
  ROOT --> INSET
  INSET --> FOOT
  SIDE --> NAV[module nav]
  INSET --> PAGE[route page]
```

FIG-06 · diagrams/06-public-shell.mmd · text/vnd.mermaid

```
flowchart TB
  HTML[own html body]
  HEAD[SHELL-PUBLIC header]
  MAIN[page]
  FOOT[tenant footer]
  HTML --> HEAD
  HTML --> MAIN
  HTML --> FOOT
  BRAND[tenant primaryColor --brand] --> HTML
```

A.3 Mermaid source FIG-07 to FIG-09

FIG-07 · diagrams/07-theming.mmd · text/vnd.mermaid

```
flowchart LR
  CLASS[".light / .dark on html"]
  VARS["--primary --sidebar --brand"]
  COOKIE[akademate-theme]
  CLASS --> VARS
  COOKIE --> CLASS
  TB[TenantBranding] --> VARS
```

FIG-08 · diagrams/08-auth-surfaces.mmd · text/vnd.mermaid

```
flowchart TB
  STAFF[SURF-DASH /auth/login]
  LEARN[SURF-CAMPUS /campus/login]
  OPS[SURF-OPS /login]
  WEB[SURF-WEB /login to app]
  STAFF --> CARD[Card Input Button]
  LEARN --> CARD
  OPS --> CARD
  WEB --> STAFF
```

FIG-09 · diagrams/09-catalog-order.mmd · text/vnd.mermaid

```
flowchart TB
  R[CAT-REPO primitives in tree]
  S[CAT-SHADCN official slug]
  M[CAT-MOTION Magic UI or Aceternity by slug]
  X[one 21st or ReUI block if pattern missing]
  R --> S
  S --> M
  M --> X
```

A.4 Mermaid source FIG-10 to FIG-12

FIG-10 · diagrams/10-design-system-delivery.mmd · text/vnd.mermaid

```
flowchart LR
  WEBDS[SURF-WEB /design-system]
  DASHDS[SURF-DASH /design-system]
  SB[Storybook :6006]
  VEND[vendor kit copy-in]
  WEBDS --> SB
  DASHDS --> SB
  VEND --> DASHDS
```

FIG-11 · diagrams/11-compose-sequence.mmd · text/vnd.mermaid

```
sequenceDiagram
  participant R as Route
  participant L as Shell
  participant T as Tokens
  participant P as Primitive
  R->>L: layout
  L->>T: CSS variables
  L->>P: Button Card Dialog
  P-->>R: page
```

FIG-12 · diagrams/12-quality-ally.mmd · text/vnd.mermaid

```
flowchart TB
  AA[WCAG AA contrast]
  FOCUS[visible focus]
  KEY[keyboard]
  SEM[semantic HTML]
  AA --> FOCUS
  FOCUS --> KEY
  KEY --> SEM
```

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